Design Assignment 2: Two-Stage OTA

Theoretical Calculations & Experimental Results

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1 Theoretical Design and Hand Calculations

The following pages contain the manual mathematical derivations for the OTA design, including:

- Circuit topology and assumed parameters.
- Transistor sizing (W/L) based on the ideal square-law model.
- Small-signal parameter (g_m, r_o) extraction and Gain (A_v) calculation.
- Frequency compensation (Pole/Zero locations and Nulling Resistor sizing).
- Transient expected values and theoretical damping factor (ζ) calculation.

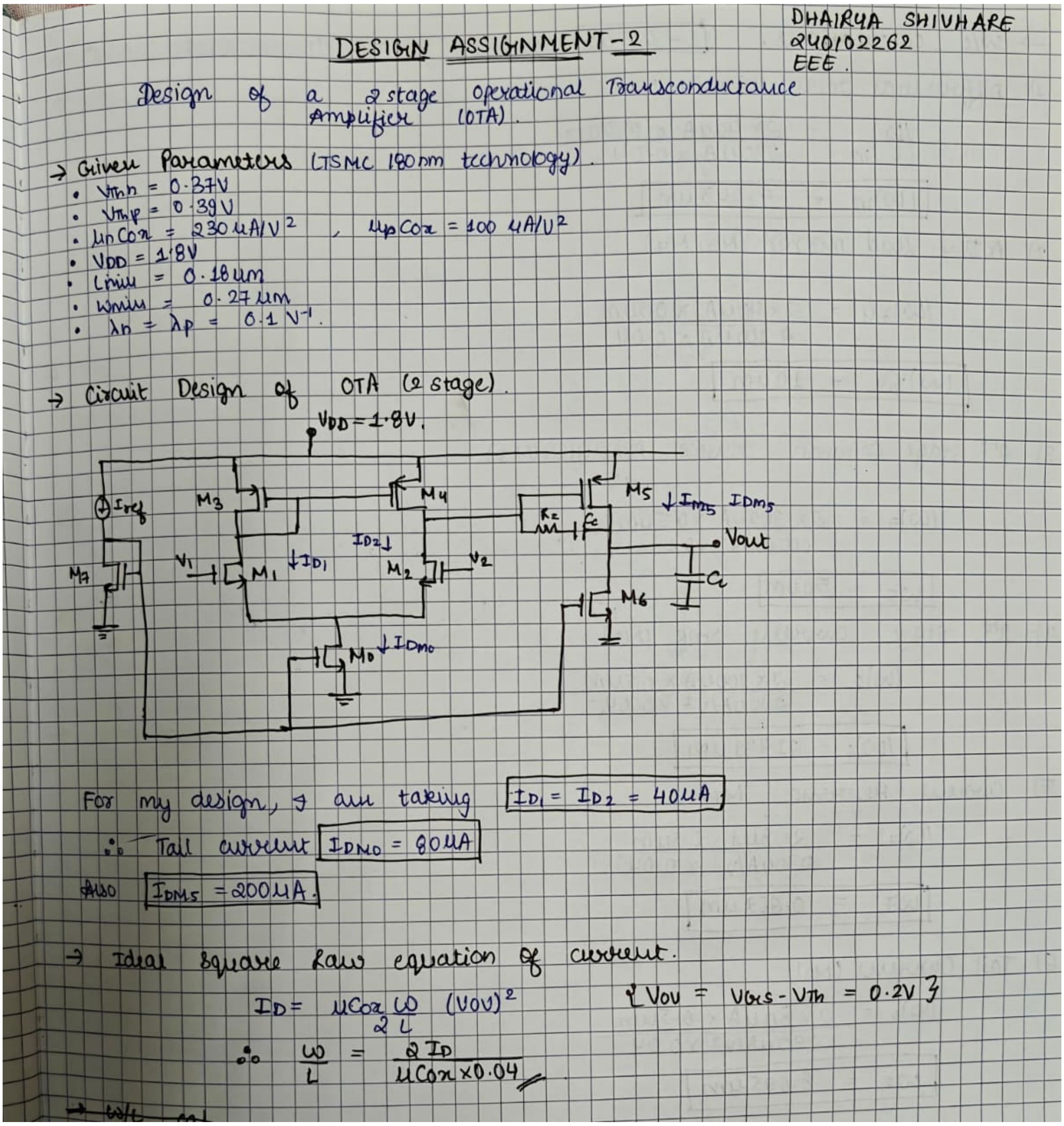


Figure 1: Given parameters and initial current budget assumptions.

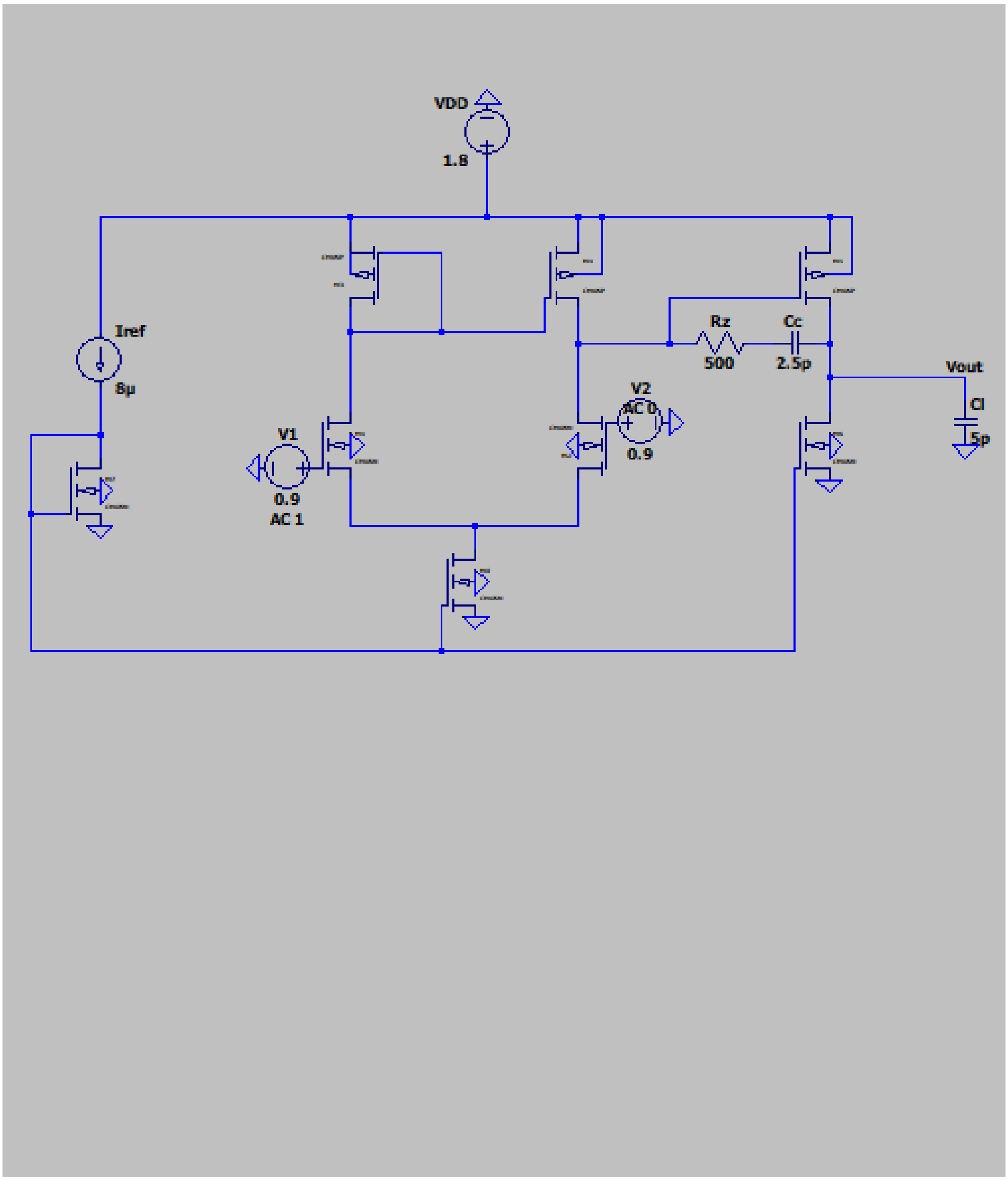
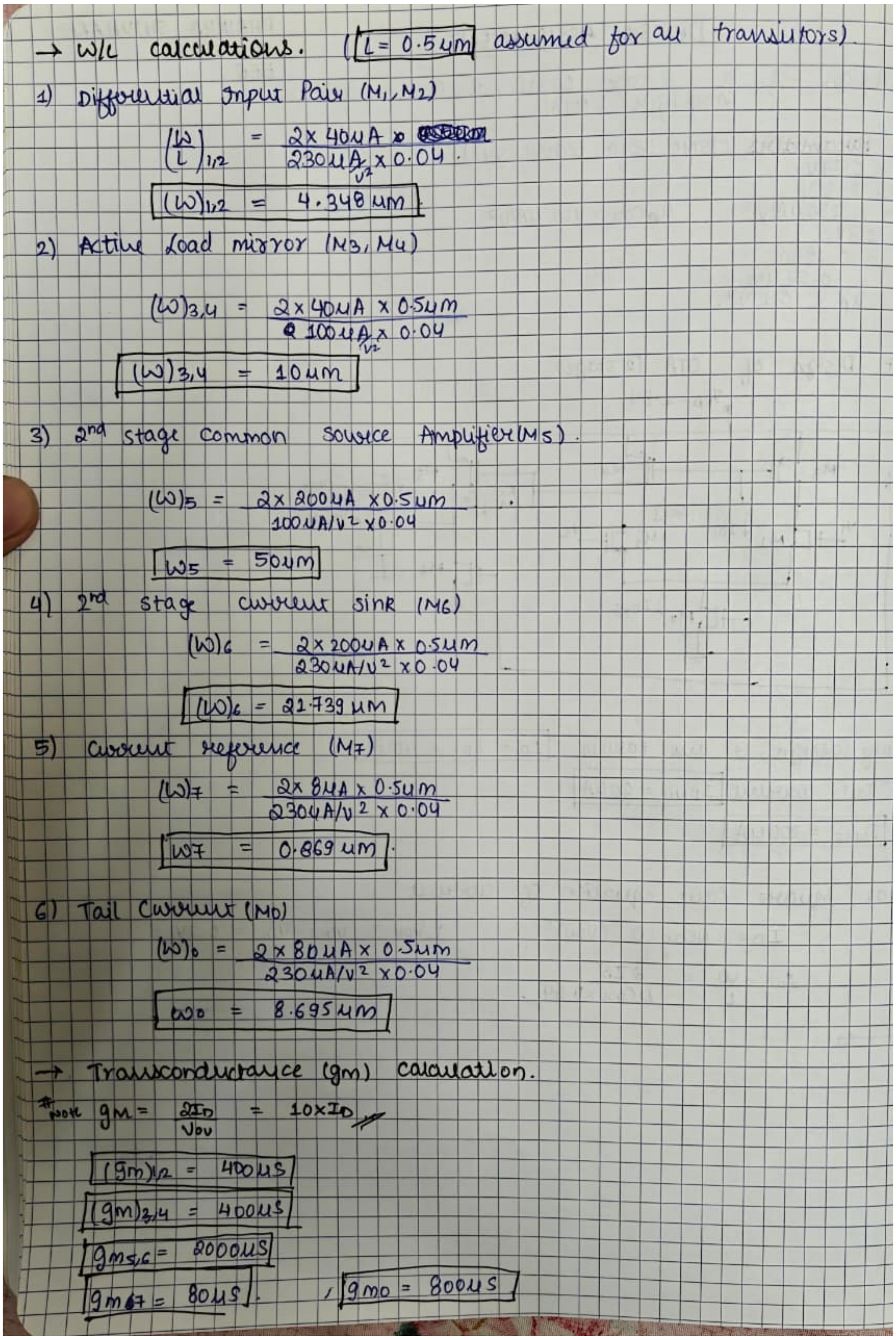


Figure 2: Hand-drawn schematic of the Two-Stage Operational Transconductance Amplifier (OTA).



→ W/L calculations. ($\boxed{L = 0.5\mu m}$ assumed for all transistors)

1) Differential Input Pair (M_1, M_2)

$$\left(\frac{W}{L}\right)_{1,2} = \frac{2 \times 40\mu A}{230\mu A/V^2 \times 0.04}$$

$$\boxed{(W)_{1,2} = 4.348\,\mu m}$$

2) Active Load mirror (M_3, M_4)

$$(W)_{3,4} = \frac{2 \times 40\mu A \times 0.5\mu m}{100\mu A/V^2 \times 0.04}$$

$$\boxed{(W)_{3,4} = 10\,\mu m}$$

3) 2nd stage common source Amplifier (M_5).

$$(W)_5 = \frac{2 \times 200\mu A \times 0.5\mu m}{100\mu A/V^2 \times 0.04}$$

$$\boxed{W_5 = 50\,\mu m}$$

4) 2nd stage current sink (M_6)

$$(W)_6 = \frac{2 \times 200\mu A \times 0.5\mu m}{230\mu A/V^2 \times 0.04}$$

$$\boxed{(W)_6 = 21.739\,\mu m}$$

5) Current reference (M_7)

$$(W)_7 = \frac{2 \times 8\mu A \times 0.5\mu m}{230\mu A/V^2 \times 0.04}$$

$$\boxed{W_7 = 0.869\,\mu m}$$

6) Tail Current (M_0)

$$(W)_0 = \frac{2 \times 80\mu A \times 0.5\mu m}{230\mu A/V^2 \times 0.04}$$

$$\boxed{W_0 = 8.695\,\mu m}$$

→ Transconductance (g_m) calculation.

note $g_m = \frac{2 I_D}{V_{ov}} = 10 \times I_D$

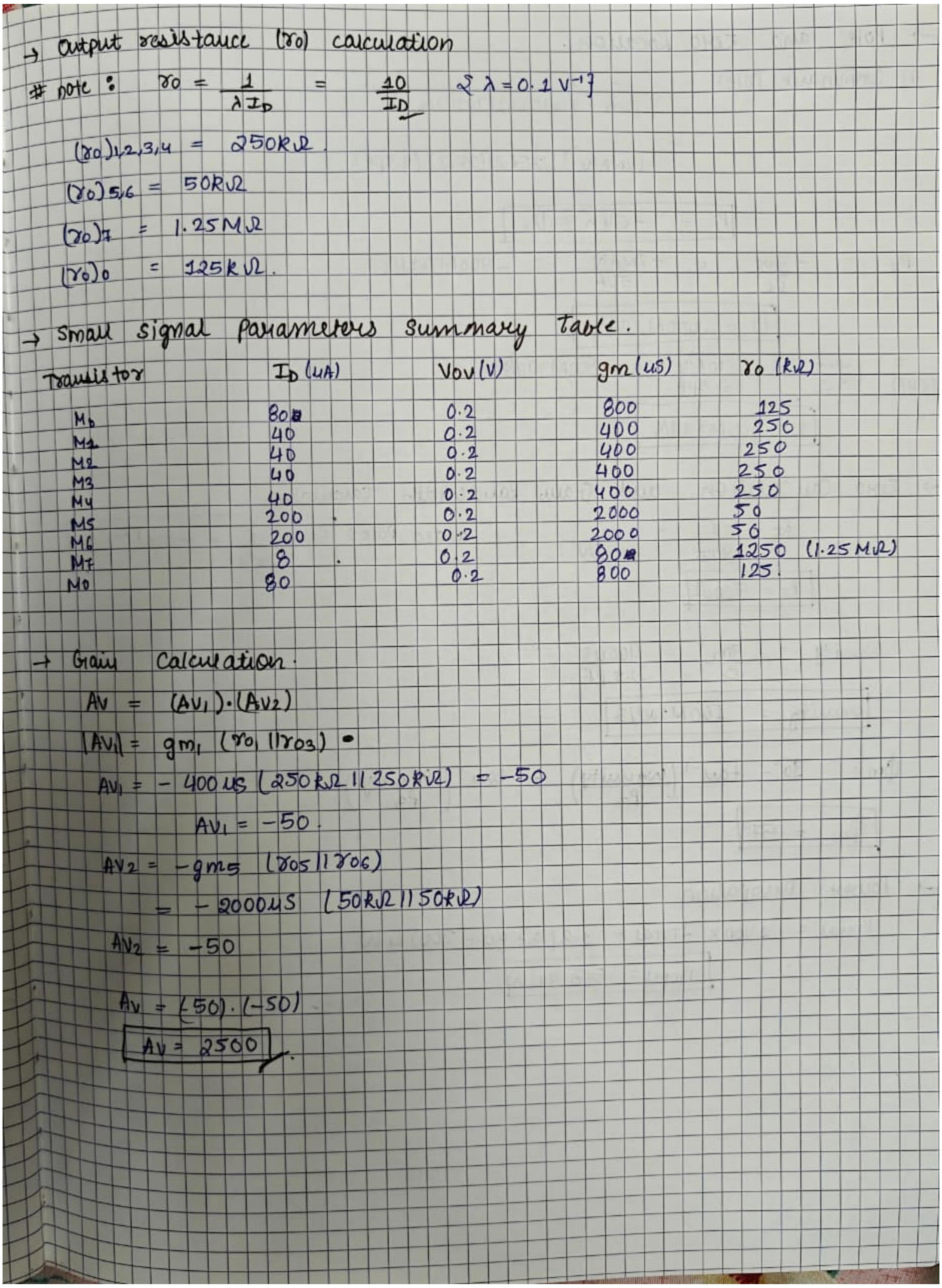
$\boxed{(g_m)_{1,2} = 400\mu S}$

$\boxed{(g_m)_{3,4} = 400\mu S}$

$\boxed{g_{m5,6} = 2000\mu S}$

$\boxed{g_{m7} = 80\mu S}$, $\boxed{g_{m0} = 800\mu S}$

Figure 3: Calculation of transistor widths (W/L) and transconductance (g_m).



→ Output resistance (r_o) calculation

note : $r_o = \frac{1}{\lambda I_D} = \frac{10}{I_D}$ $\{\lambda = 0.1\ V^{-1}\}$

$(r_o)_{1,2,3,4} = 250k\Omega$

$(r_o)_{5,6} = 50k\Omega$

$(r_o)_7 = 1.25M\Omega$

$(r_o)_0 = 125k\Omega$

→ Small signal parameters summary table.

Transistor	I_D (uA)	Vov (V)	gm (uS)	r_o (kΩ)
M0	80	0.2	800	125
M1	40	0.2	400	250
M2	40	0.2	400	250
M3	40	0.2	400	250
M4	40	0.2	400	250
M5	200	0.2	2000	50
M6	200	0.2	2000	50
M7	8	0.2	80	1250 (1.25 MΩ)
M0	80	0.2	800	125

→ Gain Calculation.

$Av = (Av_1)\cdot(Av_2)$

$|Av_1| = gm_1\ (r_{o1} \| r_{o3})$

$Av_1 = -400uS\ (250k\Omega \| 250k\Omega) = -50$

$Av_1 = -50$

$Av_2 = -gm_5\ (r_{o5} \| r_{o6})$

$= -2000uS\ (50k\Omega \| 50k\Omega)$

$Av_2 = -50$

$A_v = (-50)\cdot(-50)$

$\boxed{Av = 2500}$

Figure 4: Output resistance (r_o), small-signal summary, and Open-Loop Gain.

→ Pole and zero Location.

$$P_1 \text{ (dominant Pole)} = \frac{-1}{g_{m5}(r_{o2}||r_{o4})(r_{o5}||r_{o6})C_c}$$

$$= \frac{-1}{(2mA/V)(125k\Omega)(2k\Omega)(2.5pF)}$$

$$\boxed{P_1 = -64K \text{ rad/s}}$$

$$P_2 = \frac{-g_{m5}}{C_L} = \frac{-2mA/V}{5pF} = -400M \text{ rad/s}.$$

$$\boxed{P_2 = -400M \text{ rad/s}}$$

$$\underset{(RHP)}{z} = \frac{g_{m5}}{C_c} = \frac{2mA/V}{2.5pF} = 800M \text{ rad/s}$$

$$\boxed{z = 800M \text{ rad/s}}$$

→ Zero Cancellation and Gain Bandwidth calculation

$$R_z = \frac{1}{g_{m5}} = \frac{1}{2mA/V}$$

added Pole: $\boxed{P_3 = \frac{-1}{R_z C_c} = -800M \text{ rad/s}}$

$$\boxed{R_z = 500\Omega}$$

$$\omega_{unity} = \frac{g_{m1}}{C_c} = \frac{400\mu S}{2.5pF}$$

$$\boxed{\omega_{unity} = 160M \text{ rad/s}}$$

$$\phi_m = 90° - \tan^{-1}\left(\frac{\omega_{unity}}{P_2}\right) - \tan^{-1}\left(\frac{\omega_{unity}}{P_3}\right)$$

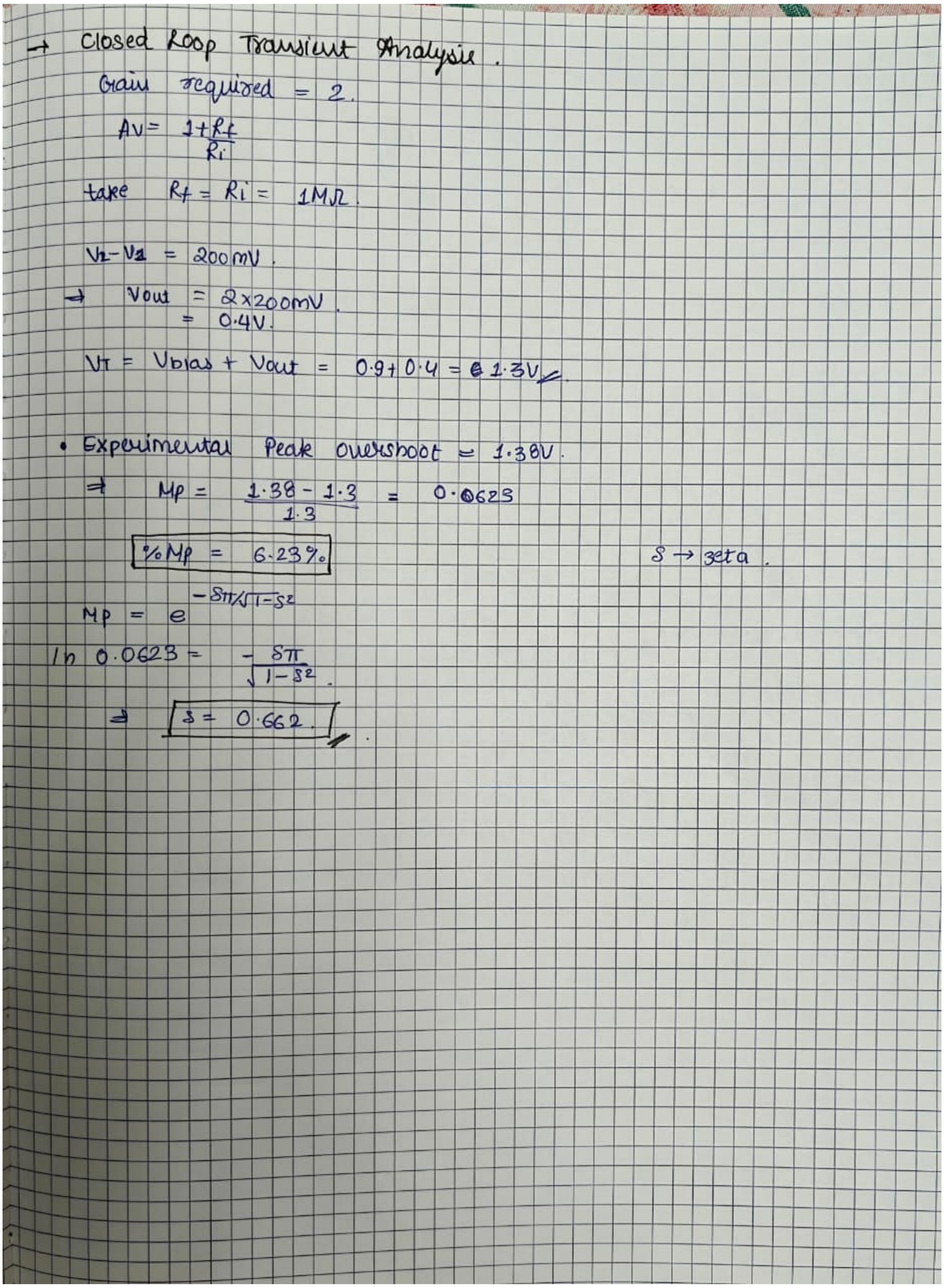
$$\boxed{\phi_m = 57°}$$

→ Power Dissipation

$$P_{Total} = V_{DD} \times I_{Total} = 1.8(8 + 80 + 200)\,\mu W$$

$$\boxed{P_{Total} = 518.4\,\mu W}$$

Figure 5: Pole/Zero location, zero cancellation, Unity Gain Bandwidth, and Power.



→ Closed Loop Transient Analysis.

Gain required = 2.

$$A_v = 1 + \frac{R_f}{R_i}$$

take $R_f = R_i = 1M\Omega$.

$V_2 - V_1 = 200mV$.

→ $V_{out} = 2 \times 200mV$.
$= 0.4V$.

$V_T = V_{bias} + V_{out} = 0.9 + 0.4 = 1.3V$

• Experimental Peak overshoot = 1.38V.

⇒ $M_p = \frac{1.38 - 1.3}{1.3} = 0.0623$

$\boxed{\%M_p = 6.23\%}$

$\zeta \rightarrow$ zeta.

$$M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

$$\ln 0.0623 = -\frac{\zeta\pi}{\sqrt{1-\zeta^2}}.$$

⇒ $\boxed{\zeta = 0.662}$.

Figure 6: Closed-loop transient analysis expected values and damping factor (ζ). Note: Hand-calculated overshoot shows a minor arithmetic rounding variant; exact simulation arithmetic follows in Section 5.

2 LTspice Circuit Schematics

The theoretical dimensions and compensation components ($C_c = 2.5\,\text{pF}$, $R_z = 500\,\Omega$, $C_L = 5\,\text{pF}$) derived in the previous section were implemented in LTspice using TSMC 180nm technology.

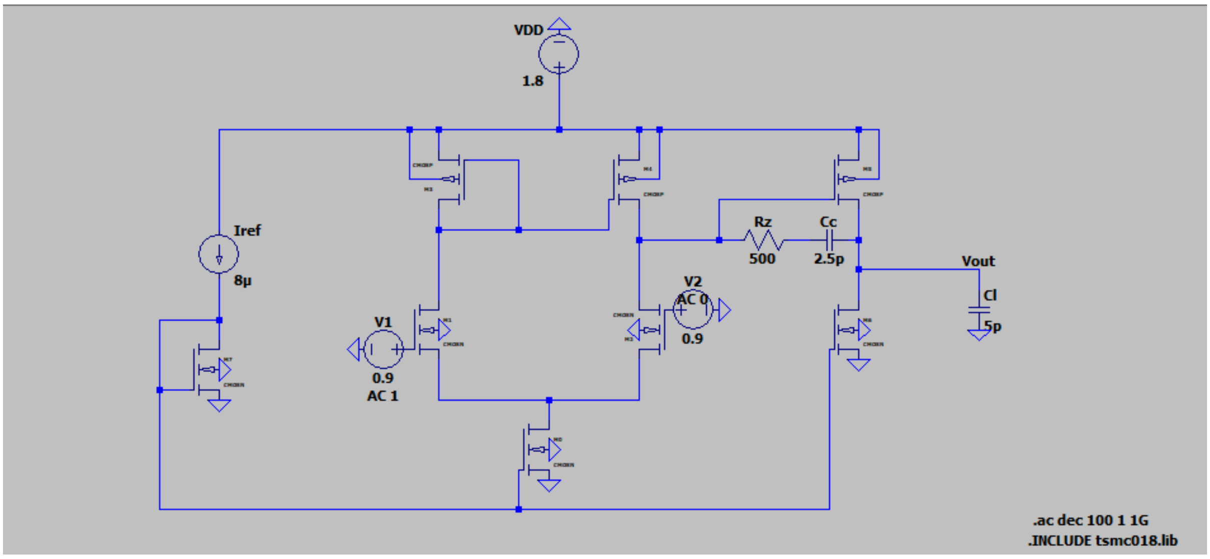


Figure 7: Open-loop AC analysis schematic configured for frequency response testing.

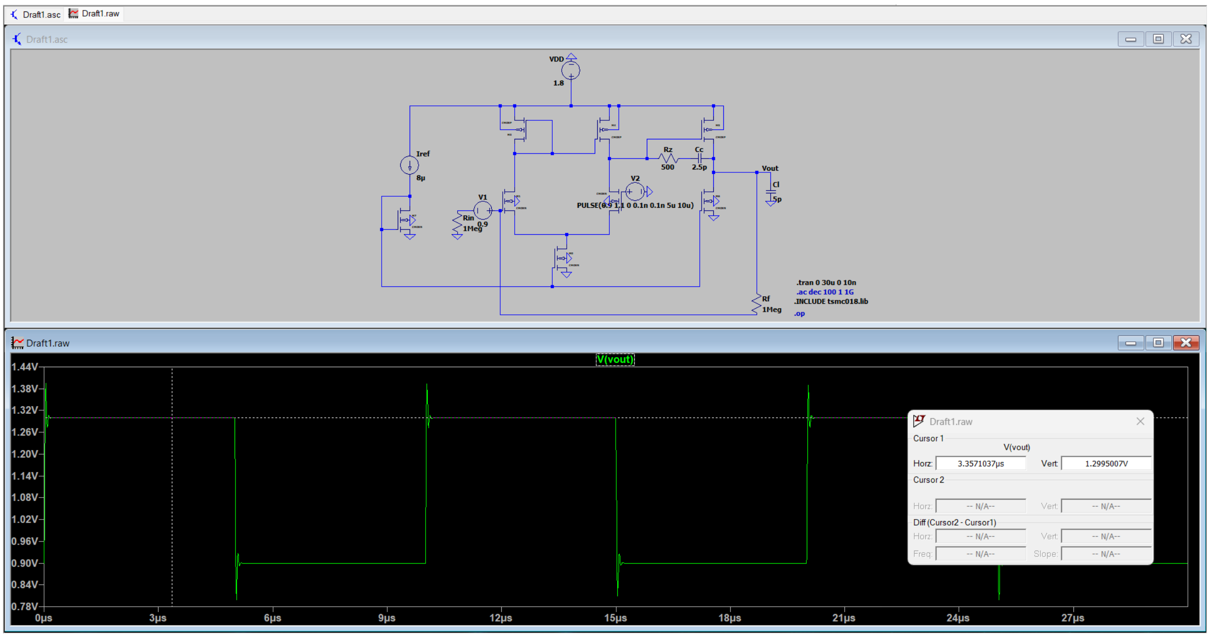


Figure 8: Closed-loop transient analysis schematic configured as a non-inverting amplifier ($A_v = 2\,\text{V/V}$).

3 DC Operating Point (.op) Validation

A DC operating point simulation was executed to verify the bias currents and extract the small-signal parameters generated by the TSMC 180nm BSIM models. The values below reflect the true operating points of the simulated circuit, accounting for real-world non-idealities such as channel-length modulation (λ) and mobility degradation.

Table 1: Simulated DC Operating Point Values (TSMC 180nm, $V_{DD} = 1.8\,\text{V}$)

Transistor	Type	Drain Current (I_D)	Transconductance (g_m)	Output Resistance (r_o)	Overdrive ($V_{GS} - V_{th}$)
M_0	NMOS	$67.05\,\mu$A	$670.5\,\mu$S	$149.1\,\text{k}\Omega$	0.2 V
M_1	NMOS	$33.52\,\mu$A	$335.2\,\mu$S	$298.3\,\text{k}\Omega$	0.2 V
M_2	NMOS	$33.52\,\mu$A	$335.2\,\mu$S	$298.3\,\text{k}\Omega$	0.2 V
M_3	PMOS	$33.52\,\mu$A	$335.2\,\mu$S	$298.3\,\text{k}\Omega$	0.2 V
M_4	PMOS	$33.52\,\mu$A	$335.2\,\mu$S	$298.3\,\text{k}\Omega$	0.2 V
M_5	PMOS	$195.59\,\mu$A	$1955.9\,\mu$S	$51.1\,\text{k}\Omega$	0.2 V
M_6	NMOS	$195.59\,\mu$A	$1955.9\,\mu$S	$51.1\,\text{k}\Omega$	0.2 V
M_7	NMOS	$8.00\,\mu$A	$80.0\,\mu$S	$1.25\,\text{M}\Omega$	0.2 V

4 Open-Loop AC Performance

An AC sweep was performed to determine the open-loop frequency response of the OTA.

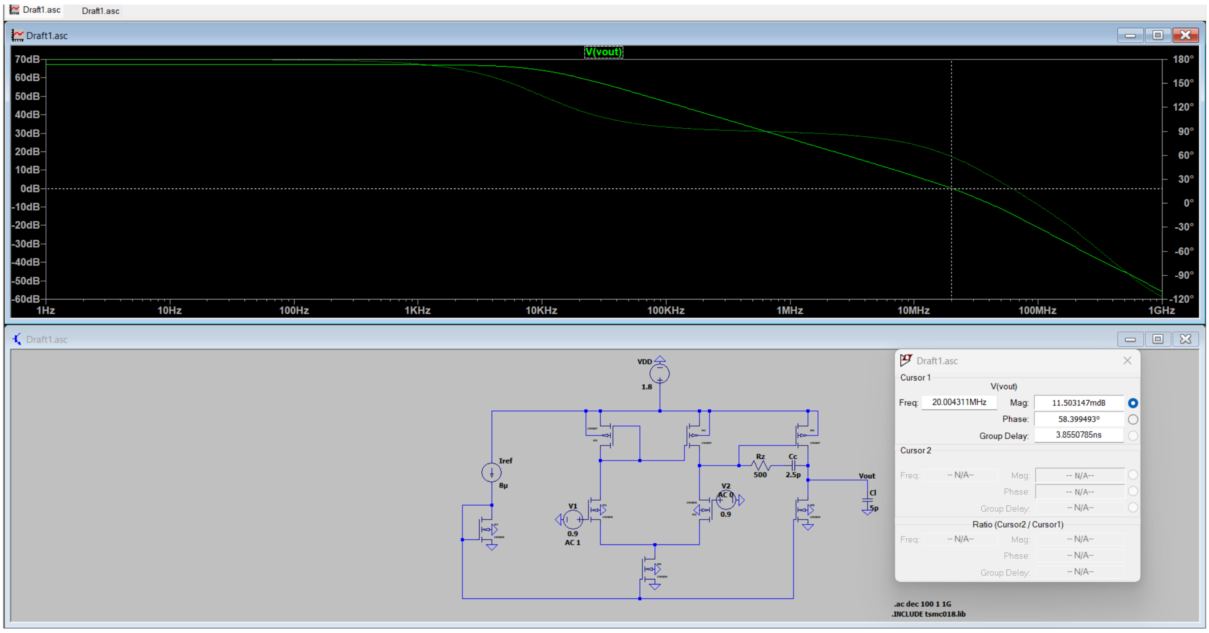


Figure 9: Bode plot demonstrating the simulated open-loop magnitude and phase response.

Experimental AC Measurements

Based on the cursor data extracted from the simulated Bode plot:

- **Simulated DC Open-Loop Gain:** $\approx 67.16\,\text{dB}$ (Exceeds the $\geq 40\,\text{dB}$ constraint, equates to a linear gain of $\approx 2300\,\text{V/V}$).
- **Phase Margin:** Extracted relative to the $-360°$ crossing at the gain crossover frequency, yielding $\phi_M = 58.39°$.

- **Unity Gain Frequency (ω_{unity}):** The simulated gain crossover frequency is extracted as $f_{unity} = 20.04\,\text{MHz}$. The experimental radial unity gain frequency is calculated as:

$$\omega_{unity} = 2\pi \cdot f_{unity} = 2\pi \cdot (20.04 \times 10^6\,\text{Hz}) \approx \mathbf{125.91}\,\text{Mrad/s} \tag{1}$$

5 Closed-Loop Transient Performance

To verify large-signal stability, the OTA was placed in a closed-loop configuration with a gain of 2 V/V. A 0.2 V step input (0.9 V to 1.1 V) was applied.

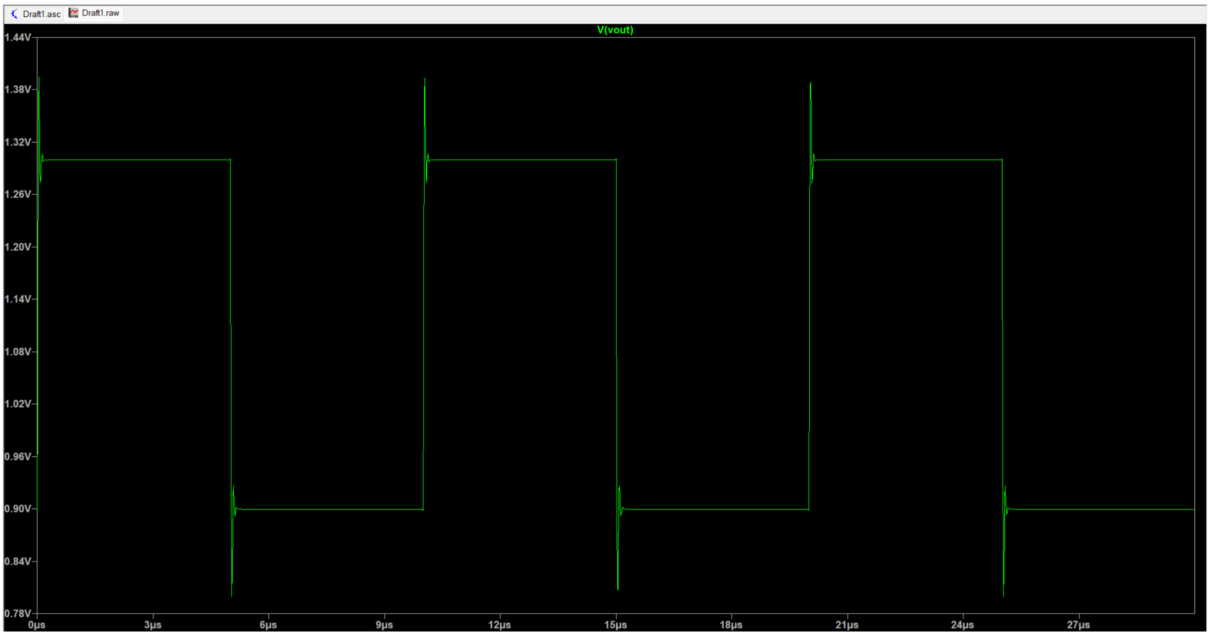


Figure 10: Closed-loop transient step response across the 5 pF load, utilizing the theoretically calculated $C_c = 2.5\,\text{pF}$.

Experimental Transient Measurements

The waveform analysis yields the following experimental data:

- **Input Step Voltage (V_{in_step}):** 0.2 V
- **Expected Output Step:** 0.4 V (Target $V_{out} = 1.3$ V DC)
- **Simulated Settled Voltage ($V_{settled}$):** 1.2995 V (Demonstrating excellent accuracy, achieving a DC closed-loop gain of 2 V/V).
- **Simulated Peak Voltage (V_{peak}):** **1.380** V

Experimental Damping Factor (ζ)

Using the simulated peak and settled voltages extracted from the cursor data, the experimental percentage overshoot (M_p) relative to the 1.3V baseline is calculated as:

$$M_p = \frac{V_{peak} - V_{settled\_target}}{V_{settled\_target}} = \frac{1.38\,\mathrm{V} - 1.3\,\mathrm{V}}{1.3\,\mathrm{V}} = 0.0615 \quad (6.15\%) \tag{2}$$

Using the standard second-order system relation $\ln(M_p) = \frac{-\zeta\pi}{\sqrt{1-\zeta^2}}$, the experimental damping ratio is:

$$\zeta_{experimental} \approx 0.663 \tag{3}$$

Overshoot is characteristic of an underdamped second-order system ($\zeta < 1$), arising from complex conjugate poles with insufficient damping — a consequence of the finite phase margin of 58.39°. This confirms that the OTA is operating in a stable but slightly underdamped regime, providing a fast response time across the 5 pF load.